



Transforming Self-regulated Learning – Multimodal Insights and Future Directions

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Abstract

Research on self-regulated learning (SRL) is undergoing a methodological and conceptual transformation from static, retrospective measures toward dynamic, multimodal, and temporally sensitive analyses. This discussion paper synthesizes and extends the contributions of a Topical Collection devoted to multimodal approaches in SRL research. It examines how diverse studies conceptualize and operationalize SRL as a complex interplay of cognitive, metacognitive, affective, and motivational (CAMP) processes. The Self-regulated learning, Multimodal data, and Analysis Grid (SMA Grid) serves as a shared framework for classifying and integrating different modalities and analytical designs. Across the reviewed contributions, a general shift from unimodal toward integrated multimodal approaches is evident, though motivational and affective dimensions remain underrepresented. The paper argues for expanding existing frameworks—particularly SMA and CAMP—toward explanatory models that account for social, contextual, and resource-based factors shaping regulatory processes. It also highlights persistent challenges in aligning data richness with theoretical depth, especially regarding temporal modeling and causal inference. A central concern is the translation of multimodal diagnostics into actionable pedagogical support, an area still underdeveloped despite the rise of AI-based analytics. Building on concepts such as cognitive load theory and resource-based perspectives, the paper proposes that SRL should be understood as a function of the dynamic balance between learners' resources, task demands, and instructional context. Ultimately, it calls for a more integrated, theory-driven, and practice-oriented research agenda that connects analysis with support, and measurement with meaningful educational intervention.

Keywords Self-regulated learning · Multimodal data · Artificial intelligence · Cognitive load · Learner resources

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Multimodal Approaches in Research on Self-regulated Learning

Research on self-regulated learning (SRL) is currently undergoing a profound conceptual and methodological transformation. In recent years, the field has moved decisively away from static, retrospective assessments toward dynamic, context-sensitive, and process-oriented approaches (Lämsä et al., 2026). This Topical Collection captures that evolution in compelling ways. It brings together theoretically grounded and empirically rigorous contributions that ask how SRL can be more precisely analyzed, more deeply understood, and more effectively supported through the use of multimodal data.

What stands out across the contributions is their theoretical and methodological depth. They explore the temporal structure of regulatory processes, situational variability, individual and social dynamics, and the growing role of new technologies — especially AI — in data generation and analysis. Together, they advance existing research programs and point toward possible directions for future work in the design and study of self-regulated learning environments.

This discussion paper builds on these contributions by offering a complementary, meta-theoretical perspective. While the individual articles advance specific empirical or conceptual foci, this discussion seeks to synthesize, reflect and extend. It aims to surface shared patterns and divergences, identify conceptual blind spots and stimulate a systematic reflection on the theoretical and methodological foundations of multimodal SRL research.

At the center of this analysis lies the SMA-Grid: the Self-regulated learning, Multimodal data and Analysis Grid (Molenaar et al., 2023), which serves throughout the Topical Collection as a classificatory and organizing framework. The first dimension of the grid is the CAMM perspective — which differentiates the cognitive, metacognitive, affective, and motivational components of SRL (Greene & Azevedo, 2010; Winne, 2015). This is also in line with current SRL models which comprise these factors and describe their interrelatedness in self-regulation (Panadero, 2017). The second dimension of the grid refers to multimodal measurements which characterize the current methodological development of SRL research (Azevedo & Gasevic, 2019). The SMA Grid constitutes the common conceptual foundation of all contributions. This discussion will critically examine this framework with respect to their explanatory power, integrative potential, and practical utility.

At the same time, the discussion considers additional factors that shape the success of self-regulated learning, such as learner resources, task characteristics and instructional contexts, and asks how such factors might be captured and supported through multimodal data.

The State of the Field: Between Progress and Fragmentation

The five contributions collected in this Topical Collection offer rich theoretical and empirical insights into how SRL unfolds as a temporally sensitive, context-dependent, and multimodally observable process. Despite their diverse methodological approaches and learning contexts, they share a commitment to capturing self-reg-

ulation not as a static trait, but as a dynamic interplay of cognitive, metacognitive, motivational, and affective processes. The following section briefly introduces each study and outlines its specific contribution to this broader endeavor.

The review by de Mooij et al. (2025) provides a broad, systematic mapping of the field along the dimensions of the SMA Grid. While previous review studies by Molenaar and colleagues (2023) focused primarily on primary studies of their own research groups, this review significantly expands the empirical basis of the SMA Grid. It identifies four types of analytical approaches: unimodal, horizontal, vertical, and integrated. This typology adds valuable granularity to the field's methodological landscape and provides a conceptual framework for comprehending diverse measurement logics. Unimodal designs focus on a single data channel — such as log files, video, or physiological signals — and analyze one CAMM dimension within that modality. Horizontal designs illuminate functional relationships among CAMM components within a single modality, while vertical designs aim at cross-modal validation of specific components. Integrated designs strive to model complex, recursive interactions across multiple modalities and multiple components, though such designs remain rare in practice.

The review documents a historical shift from unimodal to more integrated approaches. Yet, motivational and affective dimensions remain underrepresented — a structural gap that persists despite theoretical consensus on their importance (de Mooij et al., 2025). Moreover, the combination of increasingly elaborate data streams with relatively basic statistical methods reveals a methodological mismatch. The core issue is not data scarcity but the limited integration and temporal alignment of heterogeneous data types.

Another key insight is the contextual dependence of analytical strategies. Unimodal and horizontal approaches are often found in higher education contexts with large samples and individual settings, while vertical approaches are more common in collaborative learning environments. Integrated designs appear particularly diverse but are still inconsistently implemented. This suggests that context is not a mere boundary condition but a structuring factor that shapes methodological choices and research questions. While AI is frequently mentioned as a potential game changer — for example, in transcription, annotation, and pattern recognition — it is not yet systematically integrated. In fact, many applications of AI appear to enhance analytic efficiency rather than conceptual depth. de Mooij et al. (2025) thus make a substantial contribution by consolidating the SMA Grid as a reference model, while also exposing its current limitations. Their work lays the foundation for further refinement and serves as an anchor for the broader discussion that follows.

The review by Törmänen et al. (2025) contributes an important corrective to prevailing trends in SRL research by foregrounding motivation — a component that has received comparatively little attention in multimodal approaches as de Mooij et al. (2025) demonstrated. Drawing on a broad body of literature, the authors argue convincingly for a dynamic and context-sensitive conceptualization of motivation. They trace a clear development from early studies using isolated, retrospective self-reports toward more fine-grained, momentary assessments that better capture motivational fluctuations during learning. This shift represents a welcome advance in theorizing motivation as a temporally embedded process within the CAMM framework.

However, the selection of studies in the review follows a rather narrow inclusion logic: Only those works that explicitly reference established motivational theories were considered. As a result, potentially relevant studies that operationalize motivation in theoretically sound but implicit ways remain excluded (e.g. Gentner et al., 2020, 2024). This limits the scope of the synthesis and contributes to the impression that motivational research in SRL is sparser than it might actually be.

Moreover, most of the reviewed studies rely on traditional self-report measures and do not engage with multimodal or integrative designs in the sense of the SMA Grid. While this reflects the current state of the field rather than a shortcoming of the review itself, it also underscores the need for more intersectional approaches that link motivational dynamics with other CAMM processes — both across time and across modalities.

Taken together, Törmänen et al. provide an important conceptual impetus: They reclaim motivation as a key variable in SRL and call for more precise theoretical framing. Future research will need to expand this foundation by systematically connecting motivational states to cognitive, metacognitive, and affective processes — and by exploring how these interrelations unfold dynamically in complex learning environments.

Bühler et al. (2025) offer a conceptually rich and methodologically grounded contribution that bridges two research traditions that have rarely intersected: self-regulated learning and mind wandering. Their review demonstrates that mind wandering — often treated as an unwanted disruption — can also function as a resource for regulation, provided that learners possess the metacognitive awareness to recognize and strategically respond to attentional shifts. This reframing opens space for a more nuanced understanding of SRL as a process that unfolds not only through deliberate control, but also through adaptive engagement with spontaneous cognitive fluctuations.

In methodological terms, the study applies the SMA Grid to classify 134 empirical studies on mind wandering. It reveals a strong reliance on self-reports and limited engagement with vertical or integrated multimodal designs. CAMM-related constructs are often addressed, particularly cognition and metacognition, while motivational and affective dimensions remain largely absent. AI is used in only a few cases — primarily to detect patterns in data, rather than as a tool for interactive support. A significant contribution of the paper is its emphasis on the temporal granularity of both SRL and mind wandering. While SRL research often adopts macro-level perspectives, mind wandering studies tend to focus on micro-level fluctuations in attentional focus. Bühler et al. argue convincingly that this micro-temporal lens may help capture the dynamic interplay between concentration, distraction, and strategic regulation — an essential yet underexplored aspect of SRL.

By outlining how mind wandering episodes can act as both, symptoms and triggers of regulatory processes, the review opens new conceptual pathways for studying regulation not only as a goal-directed activity but also as a context-sensitive response to shifting mental states. Their work invites future research to more systematically integrate mind wandering into multimodal SRL frameworks and to develop analytic models capable of capturing this interplay across time, modality, and levels of awareness.

The empirical study by Kubsch et al. (2025) extends the field in a different direction, bringing the promise of multimodal SRL research into real-world classrooms. Their design combines affective, metacognitive, and cognitive indicators collected over multiple phases of instruction and analyzed through machine learning to predict students' productivity. This multi-phase structure enables a more nuanced understanding of how early emotional and regulatory states shape subsequent learning behaviors. Notably, the study exemplifies one of the few genuinely integrative applications of the SMA Grid within the Topical Collection.

Yet, this contribution also highlights broader conceptual tensions in the field. The use of indirect indicators such as self-reports and learning artifacts offers practical advantages, especially for scalable classroom applications.

The reliance on predictive modeling underscores another epistemological shift: the move from understanding to forecasting. While machine learning yields robust predictions, it offers limited insight into the causal or functional mechanisms underlying observed patterns. This raises an essential question: should the goal of multimodal SRL research be to recognize patterns, or to explain and ultimately support the processes those patterns reflect?

In this light, the Kubsch et al. study opens a productive line of inquiry into the role of AI not only as an observer, but as a potential co-regulator of learning (Järvelä et al., 2023; Edwards et al., 2025; Jin et al., 2023). Their findings suggest practical ways of supporting teachers, but also raise further questions: How should AI-generated insights be translated into pedagogical interventions? What kinds of scaffolding are feasible, fair, and beneficial in multimodal classrooms? The study thus points beyond itself — to a future in which analytic systems are not only diagnostic but actively supportive of learner regulation.

Finally, the contribution by Moreno et al. (2025) brings a novel perspective to the Topical Collection by foregrounding physiological and social dimensions of SRL. Drawing on data from high-fidelity medical simulation settings, the authors combine behavioral coding of individual and socially shared regulatory strategies with physiological data, specifically electrodermal activity (EDA). Their use of supervised machine learning to link regulatory behaviors to physiological arousal illustrates a powerful cross-level integration of observable and internal processes. What makes this study particularly valuable is its attention to the temporal and contextual embedding of regulation. Rather than relying solely on static markers, Moreno et al. (2025) analyze patterns across distinct phases of activity, revealing how socially shared regulation relates to subsequent emotional stabilization. This approach not only reinforces the value of temporal granularity but also highlights the regulatory function of social dynamics — a dimension largely absent from traditional CAMM perspectives (see de Mooij, 2025).

Importantly, the study positions EDA not as a predictor but as a dependent variable, shifting the analytic focus from behavior-based forecasting to physiological interpretation. This decision broadens the scope of multimodal inquiry by suggesting that internal states, such as cognitive load or stress, might serve as indicators of regulatory demand. While physiological measures are often ambiguous in isolation, their combination with behavioral data strengthens interpretability and opens new avenues for understanding learner states (Azevedo & Gasevic, 2019).

The emphasis on socially shared regulation (SSRL, Järvelä & Hadwin, 2013) also challenges the individualistic framing of SRL that underlies much of the field. By empirically modeling collective regulatory dynamics, Moreno et al. (2025) suggest a conceptual extension of CAMM — perhaps toward a CAMMS model that includes somatic and social dimensions. This theoretical expansion may be particularly relevant for collaborative or high-stakes learning environments, where regulation is co-constructed, emotionally charged, and embodied (Hadwin et al., 2017; Febriantoro et al., 2023).

Overall, the study by Moreno et al. (2025) advances both the methodological and theoretical boundaries of SRL research. It raises critical questions about the kinds of data that matter, the processes they reflect, and the frameworks we need to make sense of them. In doing so, it pushes the conversation beyond conventional modalities and toward a richer, more integrated understanding of self-regulated learning.

Taken together, the contributions paint a heterogeneous but complementary picture of current SRL research. The contributions by de Mooij et al. and Bühler et al. systematically use the SMA grid to determine positions; in the other papers, the conceptual integration remains either implicit or selective. Across the Topical Collection, CAMM components are addressed unevenly — comprehensively by de Mooij et al. and Kubsch et al., selectively by Törmänen et al. and Bühler et al., and expansively by Moreno et al.

The contributions thus stand in productive tension with one another: they significantly expand the methodological toolbox of SRL research, but at the same time reveal theoretical gaps and varying degrees of conceptual integration. The following chapters take up these tensions and discuss how viable the currently used analysis models — in particular the SMA grid — actually are, what additions could be valuable, and how a stronger connection between empirical analysis and instructional support could be achieved.

From Grid to Theory: the Explanatory Scope of SMA and CAMM

The Topical Collection proposes the SMA Grid and CAMM framework as central structuring tools for organizing and interpreting multimodal data in SRL research (Lämsä et al., 2026). These frameworks offer valuable contributions: the SMA Grid as a heuristic that visualizes methodological diversity, and CAMM as a conceptual taxonomy of core SRL components. At the same time, their application in the Topical Collection also reveals areas where further conceptual development appears necessary, particularly with regard to the integration of different process dimensions, modalities, and perspectives.

The SMA Grid serves as a valuable tool for structuring the rapidly growing field of multimodal SRL research. By positioning studies along the dimensions of modality and regulation process, it offers a shared framework that promotes transparency, comparability, and conceptual orientation.

This organizing function is productively applied in different contributions of the Topical Collection. They illustrate the grid's flexibility—from heuristic classification to guiding analytic design. At the same time, the SMA Grid does not claim to provide

a fully developed theoretical model of SRL. Rather, it supports existing frameworks by helping researchers visualize how different modalities and process components are connected in empirical studies. The grid remains open to various theoretical interpretations—a strength that allows for broad applicability, but also places the responsibility for theoretical grounding on the individual study.

For future work, it might be fruitful to align the grid more closely with dynamic process models that conceptualize SRL as a context-sensitive and interconnected system. This would not replace existing theories but strengthen the link between empirical design choices and theoretical reasoning—moving from classification toward deeper integration.

Similarly, CAMM provides a useful framework to distinguish between the domains of cognition, metacognition, affect, and motivation. Integrated approaches would allow researchers to examine the complex and dynamic interplay of these four components over time. The studies in this Topical Collection indicate a shift in this direction, yet the focus often remains limited. The reviews by de Mooij et al. and Bühler et al. highlight that motivational and affective components are frequently neglected. In contrast, the review by Törmänen et al. focuses explicitly on motivation, but does not seek to connect it with the other CAMM domains. For SRL research, it would therefore be valuable to pursue more horizontal studies that explore the interrelations among CAMM components in a systematic way. This could strengthen theoretical development in the field and leverage the potential of CAMM as a dynamic process model.

The study by Moreno et al. illustrates particularly well the intricate relationship between cognitive and affective components. Moreover, it introduces an additional dimension: social processes and dynamics in socially shared regulation of learning (SSRL). This suggests the usefulness of expanding the CAMM framework into CAMMS to account for these aspects. An expanded CAMMS model may better reflect the embodied and distributed nature of regulation in complex learning environments.

A deeper limitation, however, lies in the absence of an account of prerequisites for regulation. None of the models explicitly address the conditions under which learners are able to engage in SRL in the first place. From a resource-based perspective, regulation is not a given capacity but a function of available cognitive and motivational resources. Learners with high working memory capacity, strong prior knowledge, or well-developed strategies may be better positioned to regulate their learning. Conversely, high task complexity or extraneous load can deplete cognitive resources to the point where regulation becomes infeasible. This dynamic is reflected in a model of Seufert (2018) that integrate cognitive load theory with self-regulated learning. This model suggests that regulation is most likely to occur when learners operate within a zone of optimal challenge—where the task is demanding enough to require regulation, but not so complex as to exceed their cognitive capacity. This perspective adds an explanatory dimension to the CAMM model: regulation is not simply a process to be observed, but a phenomenon shaped by the dynamic interplay between task demands and available learner resources.

Kubsch et al. support this view by showing that prior knowledge becomes relevant later in the regulation process. Unlike classic aptitude–treatment interaction studies, where learner characteristics are treated as stable moderators used for personaliza-

tion, their findings suggest that prior knowledge should be considered repeatedly throughout the learning process—especially at points where it becomes most relevant for supporting regulation. With learning analytics approaches like those used by Kubsch and colleagues, such temporally sensitive and interconnected data collection becomes feasible.

In summary, the SMA Grid and CAMM serve as essential scaffolds for the current phase of multimodal SRL research. But as the field matures, these frameworks should evolve from descriptive tools into explanatory models. Doing so requires integrating assumptions about prerequisites (resources and load), dynamics (temporal causality), and context (social and instructional). Only then can multimodal SRL research fulfill its dual promise: to understand learning as it unfolds — and to inform interventions that support it.

Multimodal Data — Multimodal Explanations?

The contributions to this Topical Collection illustrate how richly self-regulated learning can now be observed through diverse data channels: from mouse trajectories and clickstreams to self-reports and physiological signals. Yet, this expansion of observational possibilities does not automatically translate into explanatory depth. Yet, with this diversity of data comes a fundamental question: Is it enough to document learning processes from multiple sources, or do we also need equally differentiated theories and models to interpret what the data actually mean? A recent study of Bernacki et al. (2024) offers a promising approach of cross-validation of behavioral and self-report data well-aligned with theoretical assumptions.

A first challenge lies in the implicit assumption that behavioral or physiological indicators can be directly mapped onto SRL processes. This assumption, however, is seldom interrogated (Azevedo & Gasevic, 2019). For example, what does a pause in typing signify? Is it metacognitive reflection, distraction or fatigue? Without a strong theoretical rationale linking indicator and process, such data remain ambiguous. Only by grounding data in theoretically informed models—of information processing, attentional control, or emotional and motivational regulation—can we move from raw signals to meaningful indicators, and from indicators to empirically testable hypotheses and theoretical explanations.

A promising conceptual pathway to explain self-regulated learning lies in examining the relationship between resources and cognitive load with regard to self-regulated learning. From a learning psychology perspective, not all tasks or situations are equally likely to trigger or enable SRL processes. What matters is whether cognitive demands are matched by the learner's available mental resources (Seufert, 2018). Building on cognitive load theory, three types of load can be distinguished (Sweller et al. 1998): intrinsic load (due to task complexity), extraneous load (caused by sub-optimal design or information presentation), and germane load (reflecting productive effort for learning). Self-regulation requires resources—both cognitive and motivational. These must be available and not entirely consumed by external demands or task-related load in order for planning, monitoring, or regulation to take place at all (Klepsch & Seufert, 2020).

In this sense, SRL can be seen not only as a process but as a function of systemic balance—between load and resources, between task complexity and instructional support. Multimodal data can be used to capture this balance: not only to track regulation as such, but also to detect cognitive states, mental effort, or resource usage. However, such interpretations require explicit theoretical assumptions and empirical validation.

Another key issue is the modeling of time and causality. A recurring theme throughout the Topical Collection is the need for temporally sensitive analyses. Many studies rely on snapshots or aggregated states to represent learning processes. Yet SRL unfolds over time: it follows phases, responds to contextual changes, and evolves in a cyclical rather than linear manner.

Capturing this dynamic nature requires more than vertical measurements of isolated constructs. Instead, researchers should conceptualize learning in terms of input–process–output cycles, in which factors like prior knowledge or motivation serve as inputs that shape the regulation process, whose trajectory in turn affects outcomes such as knowledge acquisition or cognitive load. These cycles are recursive: the output of one process can become the input for the next (e.g. Schmitz & Wiese, 2005). The tetrahedral model of memory research from Jenkins (2014) offers the same perspective, highlighting the importance of learner factors as well as contextual factors, such as the task itself and the environmental resources available for the learning process and outcomes. Multimodal data, in principle, offer the tools to visualize and analyze such dynamics. But this potential remains largely untapped unless temporal sequences are explicitly modeled and causal hypotheses are grounded in theory. The challenge is to move from state data to process data—and from correlations to informed assumptions about causal mechanisms.

In short, multimodality must not become a goal in itself. Its value lies in what it enables: theoretically informed, temporally sensitive, and functionally relevant explanations of learning in action.

From Analysis to Support: the Missing Link in SRL Research

While the studies in this Topical Collection offer increasingly sophisticated insights into the analysis of self-regulated learning, the translation of these insights into actionable support strategies remains underdeveloped. Only the contribution by Kubsch et al. explicitly addresses the potential for using data to inform instructional practice — and even there, the role of teachers as agents of SRL support is only tentatively explored.

This gap is particularly striking given that the SMA Grid, as formulated in the editorial framing, explicitly aims not only to improve the measurement and understanding of SRL but also to inform its support (Lämsä et al., 2026). However, only few contributions articulate how such support might be operationalized. There is little discussion of the pedagogical competencies required to interpret multimodal diagnostic data, let alone to act on it in differentiated and effective ways. Teachers, for instance, are implicitly positioned as recipients of AI-generated insights, but not as empowered co-regulators. The assumption appears to be that knowing more

about students' regulatory states will naturally lead to more effective teaching. But this overlooks a crucial mediation layer: teachers must not only access data, but also possess the pedagogical repertoire and theoretical understanding to translate it into responsive action (European Commission, 2022). A student with motivational dysregulation requires a different form of support than one struggling with metacognitive planning — and yet this level of diagnostic specificity is rarely matched by equally specific intervention strategies.

This points to a broader structural challenge in SRL research: the separation of diagnosis from support. While great effort has gone into capturing when and how learners regulate, much less attention has been paid to the design of interventions that scaffold these processes just-in-time. Moreover, most interventions remain generic rather than individualized, undermining the promise of precision that multimodal diagnostics seem to offer.

The same asymmetry appears in the use of AI. Molenaar (2022) suggests to use AI to inform different stages of instructional design in her DDA framework: to *detect* current states, to *diagnose* what the detected data mean with respect to conceptual criteria and to *act* based on the diagnose. As discussed, the current studies employ AI primarily for pattern recognition and classification — that is, for detect and diagnose stages. But the act stage remains conceptually and technically underdeveloped. What would it mean for an AI system not only to identify suboptimal regulation but to intervene meaningfully? Could AI serve as a co-regulatory partner, prompting reflection, suggesting strategies or adapting task parameters? Such scenarios require not only technical sophistication but a robust pedagogical theory of support — one that is currently lacking (Järvelä et al., 2023).

The absence of this support perspective is not merely a missed opportunity; it risks reinforcing a technocentric view in which data capture is seen as an end in itself. If SRL research is to fulfill its educational promise, it must take seriously the challenge of translation: from data to diagnosis, from diagnosis to intervention, and from intervention to meaningful learning gains. This requires a renewed focus on human–machine collaboration, teacher agency and the instructional design principles that can turn insight into impact (Ouyang et al., 2023; Zheng et al., 2024).

Meta-reflection and Outlook: from Differentiation to Integration

Taken together, the contributions in this Topical Collection offer a vivid snapshot of a field in transition. They illustrate the impressive advances in capturing SRL with multimodal data and the increasing conceptual clarity brought by frameworks such as the SMA Grid and CAMM. At the same time, they also reveal persistent challenges: a lack of explanatory models, limited integration between components with practical applications still in their infancy.

One of the most compelling opportunities for future SRL research lies in integrating a resource-oriented perspective with multimodal process analysis (Seufert et al., 2024). The studies in this Topical Collection offer rich documentation of what learners do — but they often leave open under what conditions these processes become possible. Self-regulation is not a given; it depends on the availability of cog-

nitive, metacognitive, and motivational resources. It does not occur in a vacuum, but emerges from the dynamic interplay between task demands and individual capacity (e.g. COPEs model of Winne & Hadwin, 1998). Especially under high cognitive load or in complex tasks, even well-developed regulatory skills may fail to unfold (Seufert, 2018). Incorporating a capacity-sensitive model of SRL can thus enhance our understanding of when regulation succeeds – and when it does not.

A related observation concerns the limited attention to input and output factors in SRL research. While the internal regulation process is modeled in detail, its contextual antecedents (e.g., prior knowledge, task framing) and its effects (e.g., learning gains, affective states) are often treated as peripheral. Yet, SRL unfolds in cycles (Schmitz & Wiese, 2005, Zimmerman, 2002): inputs condition the process, and outcomes in turn reshape the learner's future engagement. Embedding SRL more explicitly in such cyclical structures would allow for a deeper and more systemic understanding of regulation — not just as a process, but as a function of the conditions under which it arises and the consequences it produces. Moreover, the role of context remains under-theorized. As de Mooij et al. (2025) show, analytical strategies often co-vary with learning settings — but the implications of this alignment remain insufficiently explored. Is it merely pragmatic, or does it reflect deeper conceptual distinctions between forms of learning? Future research should more explicitly theorize how instructional design, group composition, or domain-specific knowledge structures shape the unfolding and feasibility of regulation.

The same holds for the role of AI. While the technical potential for real-time diagnosis is demonstrated, its conceptual integration as a support agent remains weak. Molenaar's detect-diagnose-act framework (2022) offers a starting point — but what would it take to build AI systems that not only observe but understand and assist learners? This question cannot be answered by engineers alone; it requires the collaborative efforts of learning scientists, cognitive psychologists, and educators.

If SRL research is to move from fragmentation to synthesis, several shifts are needed. First, it must embrace integrative, temporally sensitive models that reflect the recursive, interdependent nature of learning. Second, it must align data strategies with explanatory goals — not only measuring more, but measuring meaningfully. Third, it must strengthen its engagement with practice: by developing pedagogical theories of support, training educators to interpret data, and designing interventions that are both adaptive and actionable. This Topical Collection lays important groundwork in all of these areas. The challenge now is to build on it — by closing the loop between analysis and support, between theory and application, and between insight and action.

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